

DISCUSSION

Main Findings: The simplest possible testbed was used, here in the form of a random tanh-network (with single scalar gain g) and there was found consistent support for the criticality hypothesis as applied to memory.

Three independently coded measures of how close to chaos a network is entail an analytical spectral-radius estimate, a numerical Lyapunov exponent, and lastly a Derrida-style perturbation-spreading factor, all of which largely agree with each other to within ~ 0.02 gain units, while the autonomous-network crossing ($g_c \approx 0.97\text{--}1.06$) matches more so the classical mean-field prediction $g_c = 1$ almost exactly (Sompolinsky, Crisanti & Sommers, 1988). Furthermore, Driving the network with continuous random input shifted this boundary much higher ($g_c \approx 1.79\text{--}1.81$), which reproduces the “input suppresses chaos effect” (Rajan, Abbott & Sompolinsky, 2010). Total linear memory capacity $MC(g)$ also rises, in that it peaks at $g \approx 1.74$, and then falls, sitting just below both driven critical-gain estimates and not exactly on or past them (relating to H1, H2). The memory-capacity spectrum itself shows a storage-vs-transfer signature too: short delays remained recoverable across almost the whole gain range, while the long delays only become recoverable in a narrow window near the peak, as related to H3. The peak's location and the overall curve shape are stable across random seed and across a $5\times$ range of network size, $N \in \{100, 200, 300, 500\}$ (H4).

In Relation to Literature:

Direct precedents: The result sits inside a more or less well-established lineage. Bertschinger & Natschläger (2004) and Legenstein & Maass (2007) were the first to show that computational performance in RNN'S peaks specifically at the order-chaos boundary, and Boedecker et al. (2012) made this even more concrete for memory capacity in Echo State Networks, and additionally linked it to Lizier's information-storage measure, so that in this project's $MC(g)$ curve where rise, the sharp peak, the decline is observed, reproduces their qualitative shape essentially, but of course in a much smaller and more transparent model.

sub-critical offset: The MC peak sitting slightly on the ordered side of g_c and not than exactly at it was definitely the case, and Matzner (2017) reported the same pattern for HyperNEAT-evolved reservoirs, which always settled on the ordered side of the edge of chaos. The same qualitative offset appears, via a completely different methodology, in the biological-network literature reviewed by Roli et al. (2018), where Shmulevich et al. (2005) and Serra et al. (2007) both found that Random Boolean Network models best matched real gene-expression data (i.e yeast, HeLa cells) at a Derrida parameter slightly below 1, not precisely at the critical value, so seeing the same small, consistent offset

recur across three unrelated model classes (here being evolved neural reservoirs, gene-regulatory Boolean networks, and the hand-built rate network) is quite strong evidence that this is a real general feature of information storage near criticality.

Generality across architectures: Kanamaru, Hensch & Aihara (2023) recently extended this same basic finding to biologically structured, spiking excitatory-inhibitory cortical networks, where they tuned the E/I balance, and used a Lyapunov dimension-based edge of high-dimensional chaos criterion. That memory capacity then peaks near criticality in a different as well as more biologically constrained architecture strengthens the case.

Mechanistic account: Storage-vs-transfer patterns were observed (H3) where the direct empirical signature of the mechanism Lizier and collaborators describe (summarised in Roli et al., 2018, Section 3): ordered dynamics are primarily dominated by information storage, which then grows toward the critical line; and information transfer rises from order into chaos and peaks just past the boundary, at which point it starts actively disrupting storage. A linear memory-capacity readout probes only storage, in effect, so it is unsurprising that it peaks slightly before the point where transfer and the nonlinear scrambling of the input history begins to dominate. This then also offers an explanation for why the peak sits on the ordered side specifically, in that right at g_c , and especially past it, the same recurrent mixing that increases information transfer also increases sensitivity to the exact input history in ways a fixed linear map would not be able to exploit, as Matzner (2017) framed this as growing sensitivity to both signal and noise, with the noise sensitivity eventually dominating.

Limitations:

Limitations that should be kept in mind first is that it's a linear readout only, as the MC as defined here captured only linearly decodable memory, and information could remain present in a genuinely nonlinear form further into the chaotic regime; Boedecker et al.'s complementary use of an information-theoretic (nonlinear) storage measure suggests the true information-storage peak could sit at a slightly different location than this demonstrated linear MC peak. Also, with regards to architecture, what was used was a dense & unstructured Gaussian connectivity, but Kanamaru et al. (2023) show the same qualitative result holds for structured, or rather biologically constrained networks, but that generalization wasn't tested. It additionally has a single control parameter, yet real/engineered systems often have several parameters that jointly affect criticality; what was done is deliberately isolate one gain for interpretability, at the cost of generality. And lastly it had finite-size or sample effects, since all crossing points and peak locations are numerical estimates from finite N or finite trajectory length, and a finite number of seeds; H4 shows the qualitative picture is robust, but exact numerical values (peak at $g=1.74$ vs. $g_c=1.81$, say) should be read as approximates by all means.

Verdict:

Knowing that a deliberately minimal and fully transparent model was used, the memory capacity peaked close to but still consistently below the boundary that separates chaotic from ordered dynamics, which in turn matches both in terms of quality and quantity patterns that were reported across this range of unrelated systems in the criticality literature that was found as a reference, including evolved neural reservoirs and biological gene regulation networks. It should overall be consistent with the criticality hypothesis core claim of system poised near but not exactly at the edge of chaos to achieve a favorable trade-off between the robustness that is needed to retain information and the flexibility to process it.

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(Disclaimer: Perplexity AI was utilized for literature research, while Claude acted as an aid for the code's verifiability and general consistency)